ML Lab 501 Report: Restoring Clean Images from Noisy Images with a Pairwise MRF

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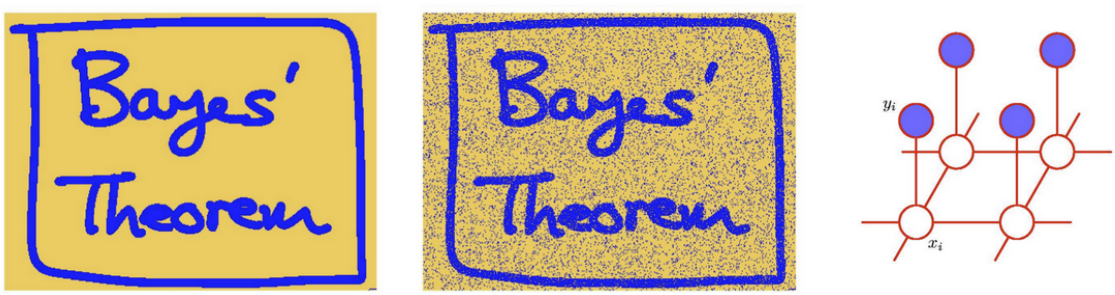
1 Overview

In this lab, a binary image corrupted by random sign flips was restored using a pairwise Markov Random Field (MRF). The goal was to recover a clean image from a noisy observation by balancing spatial smoothness and agreement with the observed pixels.

An MRF-based denoising algorithm was implemented with synchronous updates and performed a grid search over the model parameters h, β, and η. The performance was evaluated using pixel-wise recovery accuracy relative to the provided clean image and parameter were tuned to improve accuracy.



(a) Clean image (to restore) (b) Noise contaminated image (c) MRF

Figure 1: Problem Statement

2 Methods

2.1 Image preprocessing

The input image was first converted to grayscale and then thresholded at 128 to produce a binary image:

$$x = \begin{cases} +1, & \text{pixel} \geq 128, \\ -1, & \text{pixel} < 128. \end{cases}$$

Both the noisy image and the clean reference image were represented as $\{-1, +1\}$ arrays.

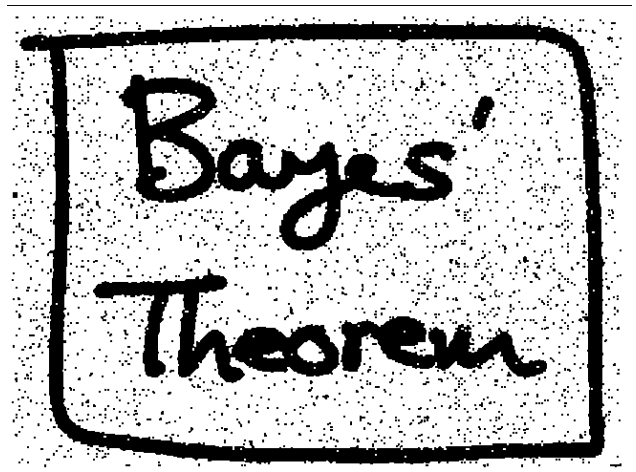


Figure 2: Greyscale image after processing.

2.2 Synchronous MRF update rule

For each pixel, the energy contribution was evaluated for assigning the pixel to $+1$ and -1, while keeping the neighboring pixels fixed. For a candidate value $x_i \in \{-1, +1\}$, the local energy contribution is

$$E_i(x_i) = hx_i - \beta \sum_{j \in \mathcal{N}(i)} x_i x_j - \eta x_i y_i,$$

where $\mathcal{N}(i)$ denotes the 4-connected neighborhood of pixel i and:

- h is a bias term that globally favors one sign,
- $\beta > 0$ encourages neighboring pixels to take the same value,
- $\eta > 0$ encourages the recovered image to stay consistent with the noisy observation.

At each iteration, the best value for every pixel was computed using the image from the previous iteration:

$$x_i^{(t+1)} = \arg \min_{x_i \in \{-1,+1\}} E_i(x_i).$$

Unlike sequential coordinate descent, this implementation updates all pixels simultaneously after computing the full next image.

2.3 Parameter Search

The three model parameters play distinct roles:

- h: shifts the global preference toward black or white pixels,
- β: controls smoothness by rewarding neighboring agreement,
- η: controls trust in the noisy observation.

The parameters were tuned over the following ranges:

$$h \in \{-1, -0.5, 0, 0.5, 1\},$$
$$\beta \in \{0.5, 1.0, 1.5, 2.0\},$$
$$\eta \in \{0.1, 0.2, \ldots, 0.9\}.$$

For each parameter combination, the algorithm was run for 2 synchronous iterations. The final denoised image was compared against the clean reference image using pixel-wise accuracy:

$$\text{Accuracy} = \frac{1}{D} \sum_{i=1}^{D} \mathbf{1}(x_i = x_i^{\text{clean}}).$$

3 Results

The best parameter combination found by the grid search was:

$$h = -1, \qquad \beta = 1, \qquad \eta = 0.5.$$

The corresponding best recovery accuracy was 97.55 %. The number of synchronous iterations used was 2 and the denoised results was shown in figure 2.

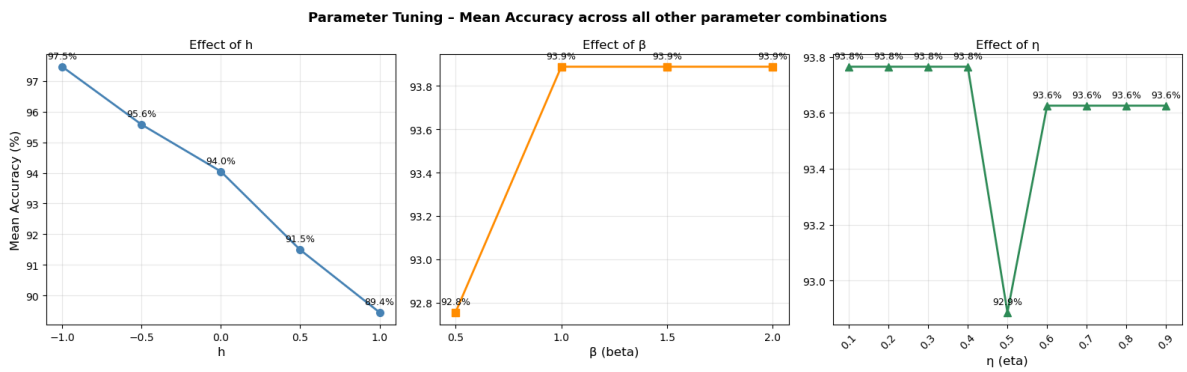


Figure 3: Mean recovery accuracy as a function of h, β, and η, averaged across all other parameter combinations.

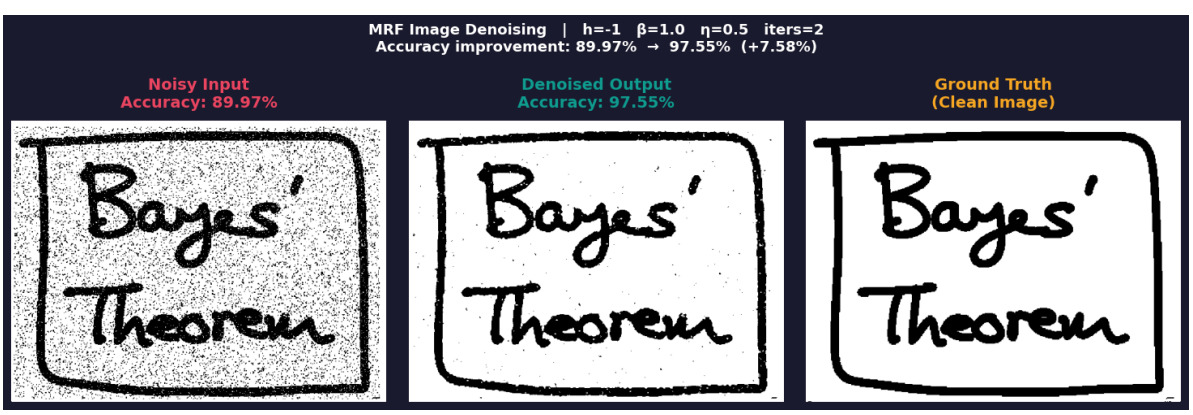


Figure 4: Example comparison between the noisy input, denoised output, and clean reference image for one parameter setting.

3.1 Qualitative observations

In general, increasing β improved local smoothness and removed isolated noise, but overly large β tended to blur fine structures. Increasing η made the solution remain closer to the noisy image; this helped preserve edges when noise was limited, but it also reduced denoising if the observation term dominated. The bias parameter h had a noticeable effect when the image contained an imbalance between black and white regions. The parameter-tuning plots also help visualize which parameters are most influential.

4 Evaluation

This model was effective because it explicitly encoded two important assumptions about natural binary images: neighboring pixels are likely to agree, and the clean image should remain reasonably close to the noisy observation. The MRF framework provided a clean way to combine these preferences into a single energy function.

The current implementation is that it uses synchronous updates for a fixed number of iterations rather than running until convergence. Because all pixels are updated simultaneously, the method may behave differently from standard coordinate descent and may be more sensitive to the chosen number of iterations. In addition, the grid search range was relatively coarse, so the best parameter setting found may not be globally optimal.

5 Reflection

This lab helped me better understand how graphical models can be used for image restoration. In particular, I learned how the balance between neighborhood smoothness and data fidelity directly affects denoising quality.

I also found that parameter tuning is essential: even when the model structure is fixed, the recovered image quality can change significantly depending on h, β, and η number of iterations. This made the connection between model design and empirical performance much more concrete.